GWS ENGINGEERING & INDUSTRIAL SUPPLIES LTD.
Link Road, Poulavone, Ballincollig, Co. Cork.
Tel: 021 4875878 Fax: 021 4875882
E: info@gwscork.ie Web: www.gwscork.ie POSTCODE: P31 YK61

CERTIFICATE OF INSPECTION (EN10204 3.1)

Certificate No.	HE/2026-2027/GW/465	**Customer Name:**	GWSCORK
Part Description:	Hex Screw M20X60 w/1N & 1PW	**Date:**	6/4/2026
Order No:	2564	**Invoice:**	HE-2345
Lot No.:	C-5327	**Part No.:**	APM R61C
Finish:	ZINC	**Size:**	M20 **Pitch** 2.5
Qty:	6000 Pcs.	**Length**	60 mm

Chemical Composition (BOLTS)

Material:	19MNB4-HC					**Specification:**		ISO-898-1					
Heat No:	SR-3792/25D03213												
Elements	C	Si	Mn	P	S	Cr	B	Mo	V				
Min	0.150	-	-	-	-	-	-	-	-				
Max	0.400	-	-	0.025	0.025	-	0.0030	-	-				
Actual:	0.300	0.220	0.990	0.020	0.008	0.430	0.0030						

Dimensional Specification (BOLTS)

Standard:	ISO 4017		**Grade:**			8.8	**Stamp:**	APM M20X60 R61C	**Type:**	After Plating		
Specification:	**Across Flat**		**Across Corner**		**Head Thickness**		**Bolt Length**					
Standard:	Min	Max	Min		Min	Max	Min	Max				
Standard:	29.67	30.00	33.53		12.28	12.71	59.40	60.60				
Actual:	29.77	29.97	34.27-34.34		12.40	12.66	59.68	60.38				

General Mechanical Characteristics of BOLTS

Specification:	ISO-898-1				
Mechanical Properties	**Standard**	**Test Values**	**Mechanical Properties**	**Standard**	**Test Values**
Hardness (HRC)	23-34	28 - 30	**Elongation (%) min.**	≥ 12	14 - 15
Tensile strength (Mpa)	≥ 830	906 - 943	**Impact strength (-20° C) (KV)**	≥ 27	60 - 64
Yield Stress (MPa)	≥ 640	655 - 685	**Torque (Nm) min.**	-	-
Proof Load (kN)	≥ 147	Pass	**Reduction of Area (%) min.**	≥ 52	54 - 55
Decarburization (mm)	≤0.015	Pass			

Dimensional Specification (NUTS)

Lot No.:			**Stamp**		APM-8SB	**Specifications**	ISO-898-2					
Standard:	ISO-4032		**Grade:**		8	**Material**	10B21		**Heat No.**	F260406		
Specification:	**Across Flat**		**Thickness**		**Across Corner**	**Element**	**C**	**Si**	**Mn**	**P**	**S**	**Cr**	**B**
Standard:	Min	Max	Min	Max	Min	Min	-	-	0.30	-	-	-	
Standard:	29.16	30.00	16.90	18.00	32.95	Max	0.580	-	-	0.048	0.058	-	
Actual:	29.76	29.94	17.46	17.88	34.37	Actual	0.214	0.100	0.856	0.016	0.004	-	

General Mechanical Characteristics of NUTS

Specification:	ISO-898-2				
Mechanical Properties	**Standard**	**Test Values**	**Mechanical Properties**	**Standard**	**Test Values**
Rockwell Hardness (HRC)	36 max.	30 - 33	**Proof Loading (KN)**	≥ 225.4	Pass
Threading	M20X2.5P	Pass	**Threading Specification:**	M20X2.5P	6H

Dimensional Specification (Plain Washers)

Standard:	ISO 7091						**Material**		Mild Steel		**Heat No.**	-	
Specification:	**Inner Dia**		**Outer Dia**		**Thickness**		C	Si	Mn	P	S	Cr	B
Standard:	Min	Max	Min	Max	Min	Max	0.240	0.150	0.450	0.042	0.039		
Standard:	22.00	22.52	35.40	37.00	2.40	3.60	**Mechanical**		**Standard**		**Test Values**		
Actual:	22.12	22.28	36.82	36.96	2.78	3.10	**Hardness (HV):**		100-200		162-172		

Surface Characteristics

Specification:	ISO 4042						
	Adherence Test		**Coating Thickness**		**ROHS**		
	Standard	**Test Values**	**Standard**	**Test Values**	**Criteria**	**Standard**	**Test Values**
Bolts	**No Peel Off**	Pass	**≥ 8 µm**	12-17	ROHS/REACH COMPLIANCE		OK
Nuts	**No Peel Off**	Pass	**≥ 8 µm**	10-15			
Washer	**No Peel Off**	Pass	**≥ 8 µm**	10-16			
Mechanical Properties		**Standard**	**Test Values**	**Specification:**			
Tensile Resistance (KN):		203.0	222 - 231				

Certified that the products have been inspected and tested in accordance with the conditions and requirements of the contract or purchase order, and unless otherwise noted, conform to the specification(s), drawing(s) relevant thereto.

We certify that the information detailed above has been reproduced from the original test certificate furnished by the manufacturer.

Niall Crowley, Quality Manager

Quality Manager